THE DIFFERENCE OF TWO THICKENINGS MORPHISMS WITH EQUAL CONORMAL MAPS IS A DERIVATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3688-3707 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3709-3732, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. First order thickenings of algebraic spaces, conormal sheaves and their functoriality, and derivations into a pushforward module; the verification is retained.

Source reference. Original lemma lemma-conormal-functorial: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 455–471. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $X \subset X'$ and $Y \subset Y'$ be two first order thickenings of algebraic spaces over B. Let $(a, a'), (b, b') : (X \subset X') \to (Y \subset Y')$ be two morphisms of thickenings over B. Assume that*

1. *$a = b$, and*
2. *the two maps $a^*\mathcal{C}_{Y/Y'} \to \mathcal{C}_{X/X'}$ (Lemma lemma-conormal-functorial) are equal.*

Then the map $(a')^\sharp - (b')^\sharp$ factors as

$$\mathcal{O}_{Y'} \to \mathcal{O}_Y \xrightarrow{D} a_*\mathcal{C}_{X/X'} \to a_*\mathcal{O}_{X'}$$

where D is an $\mathcal{O}_B$-derivation.

Proof. Instead of working on Y we work on X. The advantage is that the pullback functor a^{-1} is exact. Using (1) and (2) we obtain a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{C}_{X/X'} & \longrightarrow & \mathcal{O}_{X'} & \longrightarrow & \mathcal{O}_X & \longrightarrow & 0 \\ & & \uparrow & & \uparrow{\scriptstyle (a')^\sharp} \uparrow{\scriptstyle (b')^\sharp} & & \uparrow & & \\ 0 & \longrightarrow & a^{-1}\mathcal{C}_{Y/Y'} & \longrightarrow & a^{-1}\mathcal{O}_{Y'} & \longrightarrow & a^{-1}\mathcal{O}_Y & \longrightarrow & 0 \end{array}$$

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Now it is a general fact that in such a situation the difference of the $\mathcal{O}_B$-algebra maps $(a')^\sharp$ and $(b')^\sharp$ is an $\mathcal{O}_B$-derivation from $a^{-1}\mathcal{O}_Y$ to $\mathcal{C}_{X/X'}$. By adjointness of the functors a^{-1} and a_* this is the same thing as an $\mathcal{O}_B$-derivation from $\mathcal{O}_Y$ into $a_*\mathcal{C}_{X/X'}$. Some details omitted. $\square$